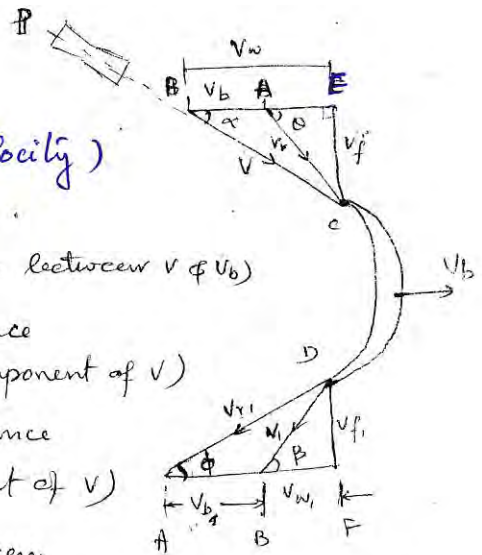


Steam turbines

Classification:

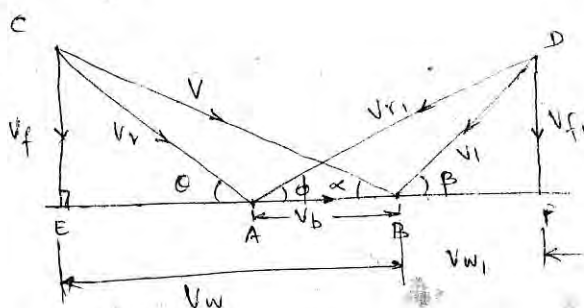
1. According to the mode of steam action
Impulse turbine, reaction turbine.
2. According to the direction of steam flow.
Axial flow turbine, Radial flow turbine
3. According to the exhaust condition
Condensing, non condensing
4. According to the pressure of steam,
High pre. ^{med (40 bar)} and low pre. (upto 2 bar)
5. According to no. of stages
Single stage multi stage.

Velocity diagram.



- V_b - velocity of moving blade.
- V - velocity of steam (abs. velocity)
- V_r - Relative velocity of the jet
(vectorial difference between V & V_b)
- V_f - velocity of flow at entrance
(vertical component of V)
- V_w - velocity of whirl at entrance
(horizontal component of V)
- α - Nozzle angle - Angle between abs. velocity and direction of motion of the blade.
- ϕ - Blade angle - Angle between relative velocity and direction of motion of the blade.

Combined velocity triangle



$$AC = AD$$

$$V_r = V_{r1}$$

PC is the axis of the nozzle, which delivers the steam jet with high velocity (V) at an angle α with the direction of motion of the blade.

The axial component of V (i.e. EC) which does no work on the blade is known as velocity of flow. It causes the steam to flow thro' the turbine, and exert axial thrust on the rotor.

The tangential component ^(BE) of V is known as velocity of whirl (V_w). The mean velocity of the blade ^(AB) is V_b .

The length AC represents the relative velocity of (V_r) of the steam jet w.r.t the blade.

The jet glides over and leaves the blade with a relative velocity V_r (DA). The absolute velocity of jet V_1 is represented by (DB) inclined at an angle β with direction of the blade motion.

The tangential component of V_1 (BF) is the velocity of whirl at exit (V_{w1}). The axial component (DF) is known as velocity of flow at exit.

$$V_r = V_{r1}$$

Force on the blade:

Let t be the time in seconds occupied by a mass M kg of steam in passing over the blade system.

Then acceleration in the direction of motion of V_b is

$$\frac{V_w - V_{w1}}{t} \quad \text{force on blade} = \frac{M}{t} (V_w - V_{w1}) = m (V_w - V_{w1})$$

But V_{w1} is opposite in motion. Hence $f = m (V_w + V_{w1})$

Work:

$W = \text{Force} \times \text{distance travelled in unit time in the direction of force}$

$$= m (V_w + V_{w1}) \cdot V_b \text{ Nm/s. or W}$$

$$\text{Power} = \frac{m (V_w + V_{w1}) V_b}{1000} \text{ kW}$$

Blade or diagram efficiency:

$$\eta_b = \frac{\text{Work done on the blades}}{\text{Energy supplied to the blades}}$$

Energy available for mechanical work = $\frac{1}{2} m V^2$

$$\eta_b = \frac{m (V_w + V_{w1}) V_b}{\frac{m V^2}{2}} = \frac{2 (V_w + V_{w1}) V_b}{V^2}$$

Axial thrust or End thrust on rotor

Axial force = Change in momentum = $m \text{ mps/sec} \times \text{Change in velocity in axial direction.}$

$$F_y = m (V_f - V_{f1})$$

When the blade friction is negligible, $V_f = V_{f1}$ and axial thrust is zero.

Effect of friction on blade:

Energy converted to heat due to friction:

$$Q_{\text{friction}} = \frac{m (V_r^2 - V_{r1}^2)}{1000} \text{ kW}$$

$$\text{Blade velocity co-efficient } k = \frac{V_{r1}}{V_r} \quad \text{(friction factor)}$$

$$\text{Stage } \eta = \text{blade } \eta \times \text{nozzle } \eta$$

$$\eta_{\text{stage}} = \frac{V_b V_w}{1000 \Delta h}$$

An impulse turbine has exit velocity of steam from a converging nozzle equal to 900 m/s. The nozzle makes an angle of 20° to the tangent of rotating wheel. The mean blade speed is 400 m/s. The blades are equiangular (symmetrical). The mass of steam flowing thro' the turbine is 1200 kg/hr. Blade velocity co-efficient $k = 0.85$. Find 1) blade angles, 2) axial thrust on blades 3) power developed.

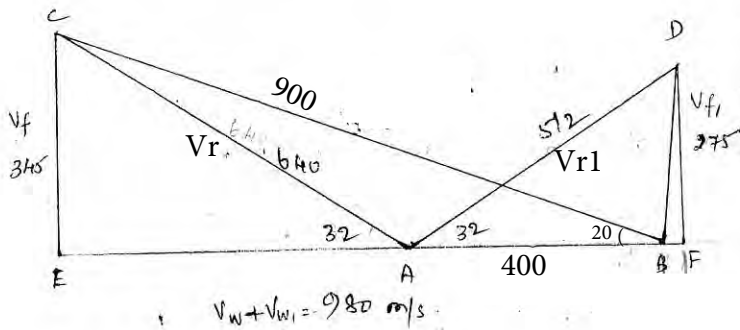
Scale of velocity diag, 1 cm = 100 m/s.

$$V = 900 \text{ m/s. } \alpha = 20^\circ \quad V_b = 400 \text{ m/s.}$$

Since blades are symmetrical, blade angle α & β are same.

Scale 1 cm = 100 m/s

$$V = 900 \text{ m/s} \quad \alpha = 20^\circ \quad V_b = 400 \text{ m/s}$$



Blade angle at inlet $\theta = 32^\circ$

$$K = 0.8 = \frac{V_{r1}}{V_r}$$

$$V_{r1} = K \cdot V_r = 0.8 \times 640 = 512 \text{ m/s}$$

$$\text{Axial thrust} = m(V_f - V_{f1}) = \frac{1200}{3600} (345 - 275) = 23.33 \text{ N}$$

Work done W_2

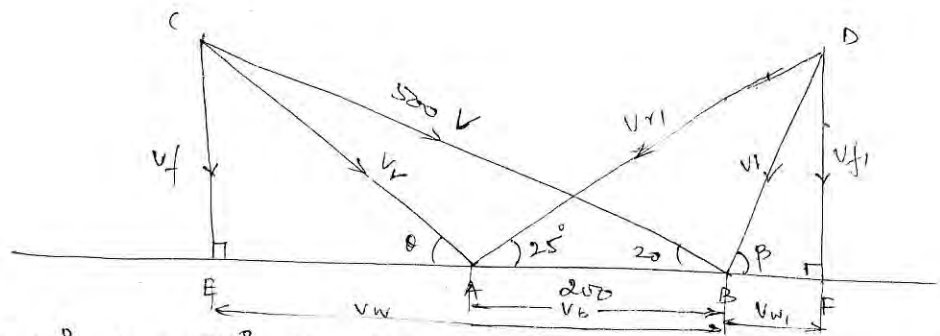
$$\text{Power developed } P = \frac{m(V_w + V_{w1}) \cdot V_b}{1000} = \frac{1200}{3600} \times \frac{980}{1000} \times 400 = 130.66 \text{ kW}$$

$$\begin{aligned} \text{Blade efficiency } \eta_b &= \frac{2(V_w + V_{w1}) V_b}{V^2} \\ &= \frac{2 \times 980 \times 400}{900^2} = 0.96 = 96.79\% \end{aligned}$$

In a De-laval turbine, the steam enters the wheel thro a nozzle with a velocity of 500 m/s. at an angle of 20° to the direction of motion of blade. The blade speed is 200 m/s and exit angle of moving blade is 25° . Find the inlet angle of moving blade, exit velocity of steam and work done per kg of steam.

$$V = 500 \text{ m/s} \quad \alpha = 20^\circ, \quad \phi = 25^\circ \quad V_b = 200 \text{ m/s}$$

$$\text{Scale: } 1 \text{ cm} = 100 \text{ m/s}$$



$$\theta = 32^\circ, \quad \phi = 59^\circ, \quad V_r = 165 \text{ m/s}, \quad V_w = BE = 470 \text{ m/s}, \quad V_{w1} = 90 \text{ m/s}.$$

$$\text{Work done} = \frac{m(V_w + V_{w1})}{1000} \text{ kW} = \frac{1(470 + 90)}{1000} = 0.56 \text{ kW}$$

$$V_w = V \cos 20^\circ = \frac{V_w}{V}, \quad V_w = V \cos 20^\circ = 500 \cdot \cos 20^\circ = 469.9 \text{ m/s}$$

$$\sin 20^\circ = \frac{V_f}{V}, \quad V_f = V \sin 20^\circ = 500 \cdot \sin 20^\circ = 171 \text{ m/s}$$

$$\tan \theta = \frac{V_f}{V_w - V_b} = \frac{171}{469.9 - 200} = 0.6335 \quad \theta = 32.35^\circ$$

$$V_r = V_{r1} = \frac{V_f}{\sin \theta} = \frac{171}{\sin 32.35^\circ} = 319.4 \text{ m/s}$$

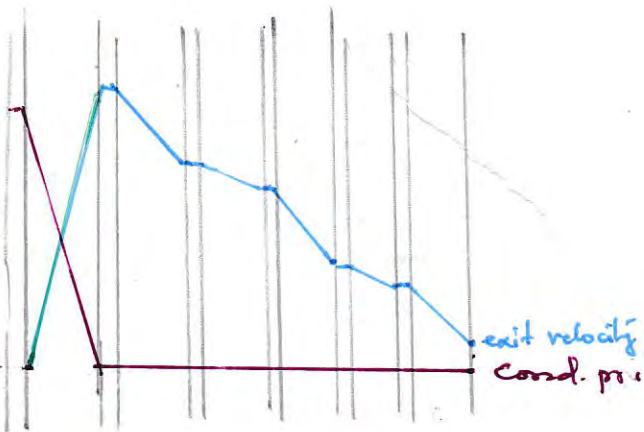
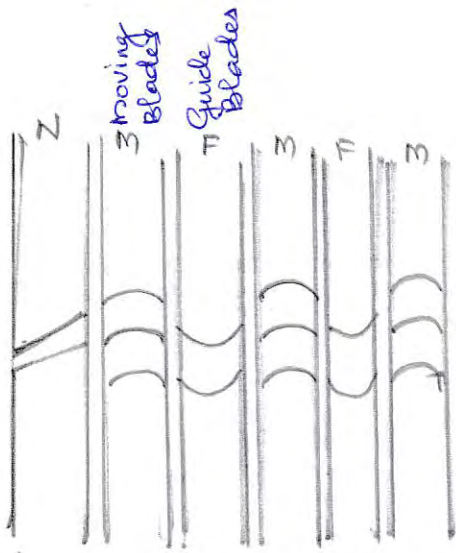
$$V_{f1} = V_{r1} \sin 25^\circ = 135 \text{ m/s}.$$

$$V_{w1} = (V_{r1} \cos 25^\circ) - 200 = 89.5 \text{ m/s}.$$

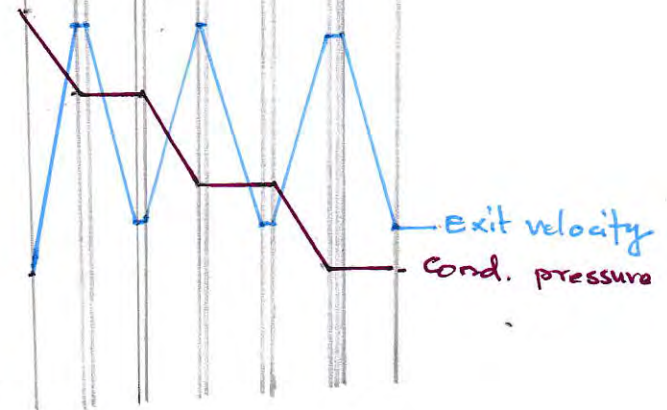
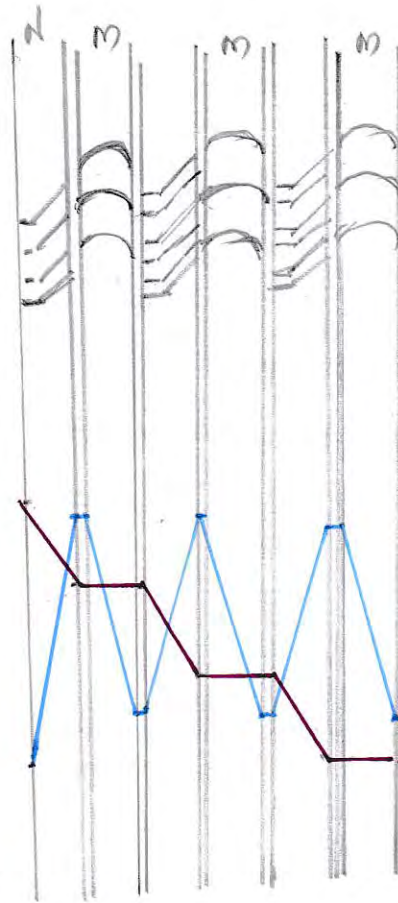
$$\tan \phi = \frac{V_{f1}}{V_{w1}} = \frac{135}{89.5}, \quad \phi = 56.47^\circ$$

$$V_1 = \frac{V_{f1}}{\sin \phi} = \frac{135}{\sin 56.47^\circ} = 162 \text{ m/s}.$$

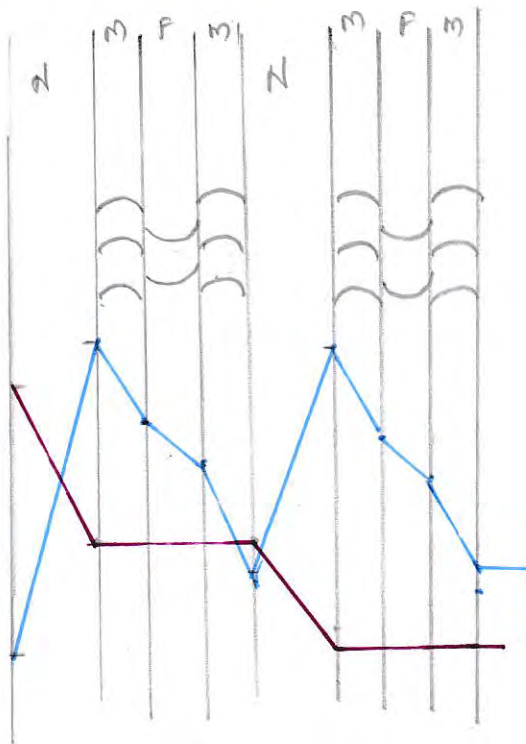
AB = 200 m/s., at B draw BC at 20° of 500 m/s.; Join AC, Draw AD at 25° with A as centre, & radius AC, draw arc on AD. Join BD, CE, DF.



Velocity compounded
Impulse Turbine



Pressure compounded
Impulse Turbine.



Pressure velocity compounding

Losses in a turbine:

- 1) Nozzle loss
- 2) Blade friction loss
- 3) ~~Mechanical~~ friction loss
- 4) Leakage loss
- 5) Moisture loss
- 6) Radiation loss
- 7) Governing loss
- 8) Residual velocity loss.

Steam Turbine Classification

Superheated steam is generated from steam boiler then to be distributed to steam turbine generator. Steam turbine has some classifications. The following below is the classification of steam turbine:

1. Based on steam flow direction

Axial steam turbine, the steam turbine which has steam flow direction parallels to the axis of shaft.

Radial steam turbine, the steam turbine which has direction of steam flow perpendicular to the axis of shaft.

2. Based on working principal

a. **Impulse steam turbine**, steam turbine which the rotation of its blades is caused by the steam where the velocity of steam had been increased by nozzle. The impulse steam turbine include

- De-Laval steam turbine
- Curtis steam turbine
- Zoelly/Rateau steam turbine
- Parson steam turbine

b. **Reaction steam turbine**, the steam turbine which the rotation of its blades is caused by reaction of its blades due to steam flow itself.

3. Based on ~~exit~~ steam exhaust

- a. Back pressure steam turbine
- b. Direct condensation steam turbine
- c. Back pressure extraction steam turbine
- d. Condensation extraction steam turbine
- e. Non condensing steam turbine with direct flow

Condensing & Non condensing

4. Based on steam pressure

- a. **Low pressure steam turbine**, the turbine with pressure up to 2 ata.
- b. **Middle pressure steam turbine**, the turbine with pressure up to 40 ata.
- c. **High pressure steam turbine**, the turbine with pressure 40 – 170 ata.
- d. **Very high steam turbine**, the turbine with pressure exceeds 170 ata. *-225*
- e. **Super critical pressure steam turbine**, the turbine with pressure exceeds 225 ata.

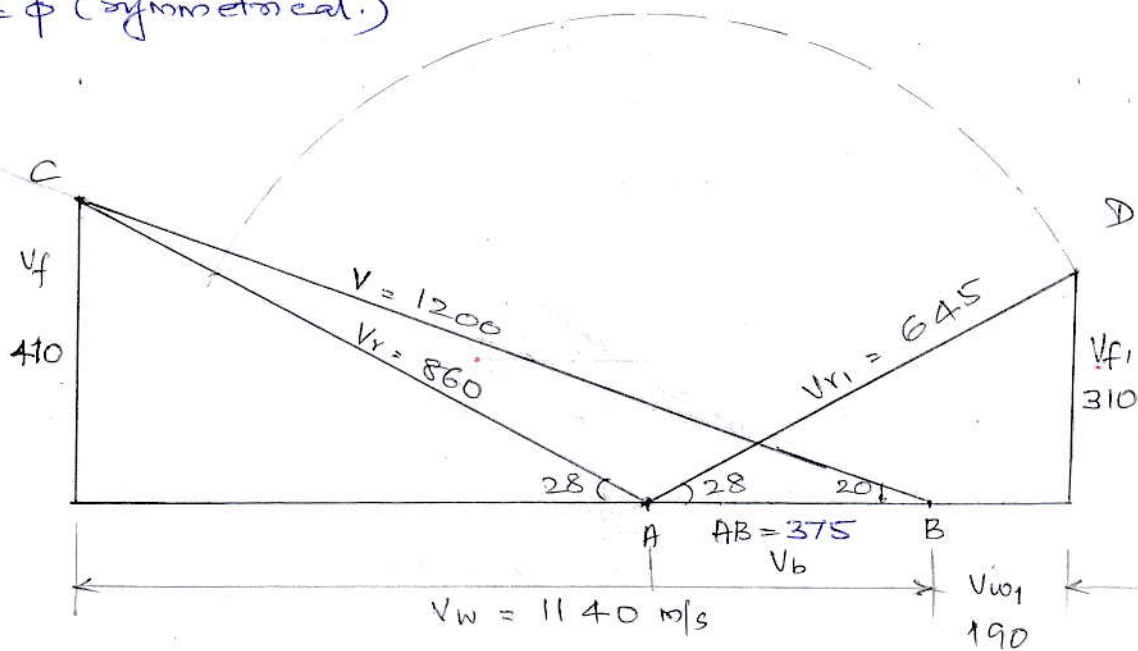
Advantages:

Higher speed, higher efficiency, less steam consumption, less space, lighter foundation, less vibration, less frictional losses, safe (enclosed in a casing)

P. An impulse turbine has exit velocity of steam from a converging nozzle equal to 1200 m/s. The nozzle makes an angle of 20° to the tangent of rotating wheel. The mean blade velocity is 375 m/s. The blades are symmetrical. The mass of steam flowing through the turbine is 0.5 kg/s. The blade velocity co-efficient is 0.75. Find 1) Blade angles 2) Force on the wheel 3) Axial thrust 4) power developed.

Scale of the velocity diagram 1 cm = 100 m/s.

$V = 1200 \text{ m/s}$, $\alpha = 20^\circ$, $V_b = 375 \text{ m/s}$, $K = \frac{V_{r1}}{V_r} = 0.75$, $m = 0.5 \text{ kg/s}$
 $\theta = \phi$ (symmetrical.)



$\theta = \phi = 28^\circ$, $V_w = 1140 \text{ m/s}$, $V_{w1} = 190 \text{ m/s}$

$V_f = 410 \text{ m/s}$, $V_{f1} = 310 \text{ m/s}$.

1) Blade angle $\theta = \underline{28^\circ}$

2) Force on the wheel $F_x = m(V_w + V_{w1})$
 $= 0.5(1140 + 190)$
 $= \underline{665 \text{ N}}$

3) Axial Thrust on the wheel.

$$F_y = m (V_f - V_{f_1}) = 0.5(410 - 310) \\ = \underline{50 \text{ N}}$$

4) Power developed. $P = m (V_w + V_{w_1}) V_b$

$$= 0.5(1140 + 190) 375 = 249375 \text{ W} \\ = \underline{249.38 \text{ kW}}$$

→ x

Without graphical method:

$$V_w = V \cos 20 = 1200 \times \cos 20 = 1128 \text{ m/s.}$$

$$V_f = V \sin 20 = 1200 \times \sin 20 = 410.4 \text{ m/s.}$$

$$\tan \theta = \frac{V_f}{V_w - V_b} = \frac{410.4}{1128 - 375} = 0.54, \quad \theta = 28.6^\circ$$

$$V_r = \frac{V_f}{\sin \theta} = \frac{410.4}{\sin 28.6} = 857.35 \text{ m/s}$$

Without graphical method.

$$V_w = V \cos \alpha = 1200 \cos 20 = 1127.63 \text{ m/s}$$

$$V_f = V \sin \alpha = 1200 \sin 20 = 410.42 \text{ m/s}$$

$$\tan \theta = \frac{V_f}{V_w - V_b} = \frac{410.42}{1127.63 - 375} = 0.5453$$

$$\theta = 28.6^\circ$$

$$V_r = \frac{V_f}{\sin \theta} = \frac{410.42}{\sin 28.6} = 857.38 \text{ m/s.}$$

$$V_{r_1} = 0.75 \times 857.38 = 643.03 \text{ m/s}$$

$$V_{f_1} = V_{r_1} \sin \theta = 643.03 \sin 28.6 = 307.82 \text{ m/s}$$

$$V_{w_1} = (V_{r_1} \cos 28.6) - 375 = (643.03 \cos 28.6) - 375 \\ = 189.57 \text{ m/s}$$